

Hybrid Energy-Based and Normalizing Flow Models for Trustworthy LLM Evaluation

Infinity Zero

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Problem Motivation

- Large Language Models (LLMs) generate fluent text but:
 - Produce hallucinated answers
 - Show overconfidence in incorrect outputs
- Existing evaluation metrics:
 - Accuracy $\rightarrow$ ignores uncertainty
 - Softmax confidence $\rightarrow$ misleading
- **Key Challenge:**
 - How to measure *trustworthiness* instead of just correctness?

Objective

- Develop a system to:
 - Detect hallucinated responses
 - Estimate uncertainty in outputs
- Assign a **Trust Score**:
 - High $\rightarrow$ Reliable response
 - Low $\rightarrow$ Hallucinated / uncertain
- Approach:
 - Combine **Energy-Based Models (EBM)**
 - Combine **Normalizing Flows (NF)**

Why Hybrid Model?

Limitations

- EBM:
 - Good classification
 - No probabilistic interpretation
- NF:
 - Models probability distribution
 - Weak classification performance

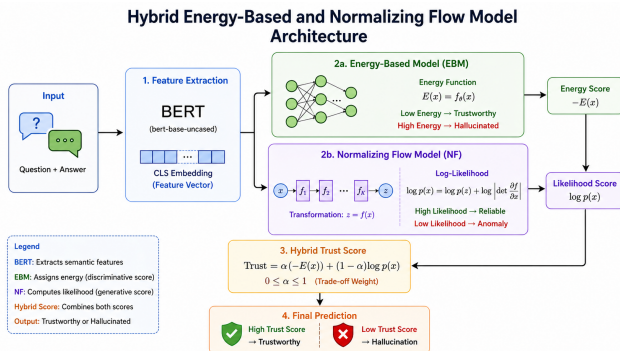
Our Idea

Combine:

- Discriminative power (EBM)
- Generative uncertainty (NF)

System Architecture

- Input: Question + Answer pair
- Feature extraction using BERT
- Parallel branches:
 - EBM $\rightarrow$ Energy Score
 - NF $\rightarrow$ Likelihood Score
- Combine scores $\rightarrow$ Final Trust Score



Energy-Based Model

- Energy Function:

$$E(x) = f_{\theta}(x)$$

- Interpretation:

- Low energy $\rightarrow$ trustworthy
- High energy $\rightarrow$ hallucinated

- Training:

- Contrastive loss:

$$L = E(x_{pos}) - E(x_{neg})$$

- Strength:

- Strong discriminative capability

Normalizing Flow

- Maps input to latent space:

$$z = f(x)$$

- Likelihood computation:

$$\log p(x) = \log p(z) + \log \left| \det \frac{\partial f}{\partial x} \right|$$

- Interpretation:
 - High likelihood $\rightarrow$ normal / reliable
 - Low likelihood $\rightarrow$ anomaly
- Models used:
 - RealNVP
 - MAF

$$\text{Trust} = \alpha(-E(x)) + (1 - \alpha) \log p(x)$$

- α controls contribution
- Combines:
 - EBM (classification)
 - NF (uncertainty)

Dataset and Preparation

- Dataset: TruthfulQA
- Total samples: 790
- Data construction:
 - Question + Correct Answer $\rightarrow$ Label 0
 - Question + Incorrect Answer $\rightarrow$ Label 1
- Task:
 - Binary classification:
 - Trustworthy vs Hallucinated

EBM Only

- Accuracy : 74.25%
- F1 Score : 0.818
- AUROC : 0.861
- Balanced Accuracy : 0.776
- Precision : 0.94
- Confusion Matrix :

$$\begin{bmatrix} 99 & 20 \\ 136 & 351 \end{bmatrix}$$

Interpretation

- Very high precision (0.94) → predictions are highly reliable
- Lower recall → misses some hallucinated outputs
- Good balance between truthful and hallucinated detection

Normalizing Flow Performance Analysis

NF Only (RealNVP)

- Accuracy : 75.24%
- F1 Score : 0.833
- AUROC : 0.777
- Balanced Accuracy : 0.731
- Precision : 0.911
- Confusion Matrix :

$$\begin{bmatrix} 83 & 36 \\ 114 & 373 \end{bmatrix}$$

NF Only (MAF)

- Accuracy : 72.77%
- F1 Score : 0.818
- AUROC : 0.740
- Balanced Accuracy : 0.668
- Precision : 0.879
- Confusion Matrix :

$$\begin{bmatrix} 68 & 51 \\ 114 & 373 \end{bmatrix}$$

Key Observations

- RealNVP performs better than MAF
- High precision but weaker AUROC
- NF struggles in classification tasks

Hybrid Model Analysis

Hybrid (RealNVP)

- Accuracy : 74.25%
- F1 Score : 0.8181
- AUROC : 0.8613
- Precision : 0.946
- Best $\alpha = 1.0$
- Confusion Matrix :

$$\begin{bmatrix} 99 & 20 \\ 136 & 351 \end{bmatrix}$$

Observation

- Fully depends on EBM
- NF contribution negligible

Hybrid (MAF)

- Accuracy: 74.25%
- F1 Score: 0.819
- AUROC: 0.862
- Precision: 0.943
- Best $\alpha = 0.65$
- Confusion Matrix :

$$\begin{bmatrix} 98 & 21 \\ 135 & 352 \end{bmatrix}$$

Observation

- Slight improvement over EBM
- NF contributes partially

Fusion Model (Best Results)

Model	Accuracy	F1	AUROC	Precision
Fusion (RealNVP)	82.84%	0.887	0.885	0.942
Fusion (MAF)	80.36%	0.868	0.889	0.946

Fusion (RealNVP) **Fusion (MAF)**

$$\begin{bmatrix} 94 & 25 \\ 79 & 408 \end{bmatrix}$$
$$\begin{bmatrix} 97 & 22 \\ 97 & 390 \end{bmatrix}$$

- Best overall performance
- Balanced detection of both classes

Conclusion

- Proposed a hybrid trust evaluation system
- Successfully detects hallucinated outputs
- Combines:
 - Discriminative + Generative modeling
- Achieved strong performance:
 - Accuracy: 82.84%
 - AUROC: 0.885

Future Work

- Handle class imbalance
- Improve NF architectures
- End-to-end joint training
- Better fusion strategies
- Real-time deployment